Chained Indices Unchained:
Structural Transformation and the Welfare Foundations of Income Growth Measurement

Omar Licandro
University of Leicester and BSE

Juan Ignacio Vizcaino
University of Nottingham

Plan

Motivation and Research Question

- We study how to measure **welfare-relevant growth** in economies undergoing:
  - structural transformation (ST),
  - investment-specific technical change (ISTC).

- Why does this matter? ST and ISTC lead to permanent changes in:
  - relative prices,
  - expenditure shares.

  Which can generate aggregation biases in traditional quantity indices.

- In this context, we ask:
  - Do chained indices provide a welfare-based measure of real income growth?
  - Are they preferred to fixed-based equivalent variation measures?

Our Approach

- Drivers of ST
  - Non-homothetic preferences: income effects.
  - Heterogeneous sectoral productivity growth: changes in relative prices.
- ISTC leads to
  - Permanent decline in the relative price of investment.
  - Affecting dynamic consumption-investment choices (Bellman Equation).
  - Leading to time-varying consumption-investment indifference maps.
- To deal with time-varying indifference maps, we apply the Fisher–Shell Principle:
  - Welfare comparisons must use a consistent preference order.
  - The natural choice is current preferences.

Our Approach

- Alternative applications of the Fisher-Shell (FS) Principle:
  - **Fixed-Base FS Indices**
    - All past is compared to the present using current preferences and prices
    - Baqaee and Burstein (QJE, 2024)
  - **Chained FS Indices**
    - Measure infinitesimal welfare gains holding local preferences and prices fixed.
    - Then chaining these gains over time.
- Following Durán and Licandro (EJ, 2025)
  - The Chained FS quantity index coincides with the Chained Divisia index in continuous time.
  - Growth rates and expenditure shares are sufficient statistics of welfare gains.

Main Findings

- Within the framework of Herrendorf, Rogerson, and Valentinyi (2021), featuring
  - non-homothetic preferences,
  - sector-specific productivity growth,
  - investment-specific technical change.
- We compare
  - The Chained Divisia Index.
  - A Current-Base Fisher–Shell Index (CB).
- Quantitative Findings
  - The CB Index substantially understates past welfare growth.
  - Historical growth is revised downwards as the base period moves.
- Information Requirement
  - The Divisia Index only requires expenditure shares and growth rates.
  - The CB Index additionally requires knowledge of deep model parameters.

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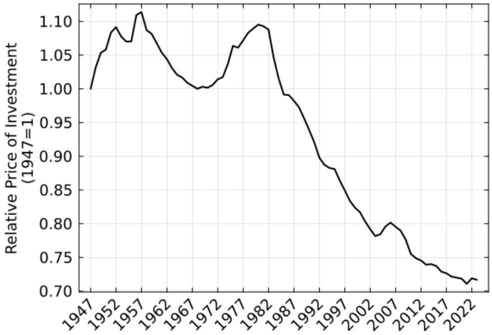
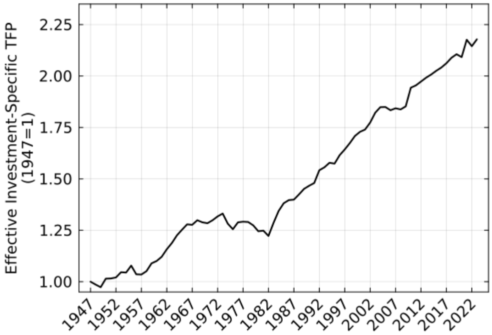
ST and IBTC

Relative Price of Investment and Aggregate TFP

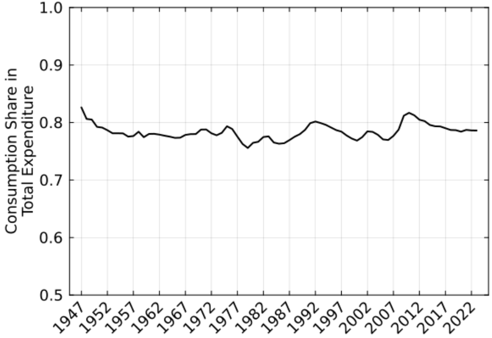


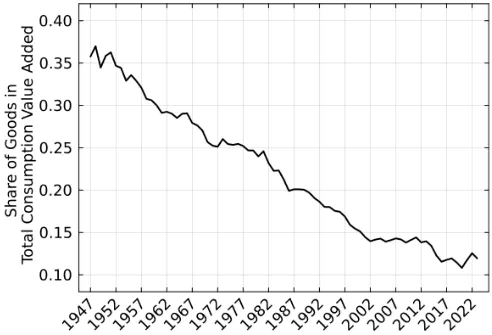


- Since 1980:
  - declining relative price of investment.
  - rising aggregate TFP, at constant rate.

ST and IBSC

Total and Consumption Expenditure Composition





- Consumption expenditure roughly stable as a share of GDP.
- Strong decline in the share of goods in total consumption expenditure.

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Structural Transformation Model

Sectoral Production

- Two consumption sectors: goods g and services s.
- The final good can be allocated to consumption expenditure or investment.
- Sectoral value added is produced with Cobb–Douglas technologies:

$$Y_{j,t} = A_{j,t} K_{j,t}^{\theta} L_{j,t}^{1-\theta}, \qquad j \in \{g, s\}.$$

- Capital and labor are fully mobile across sectors,
- $+$ firms face common equilibrium factor prices (r_t, w_t):

$$\rightarrow \frac{K_{g,t}}{L_{g,t}} = \frac{K_{s,t}}{L_{s,t}}$$

Structural Transformation Model

Investment Technology

- Investment is produced using goods and services as intermediate inputs:

$$I_t = A_{x,t}\left[\omega^{1/\varepsilon} X_{g,t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} X_{s,t}^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}.$$

- $X_{g,t}$ and $X_{s,t}$ are goods and services used in investment production.
- $A_{x,t}$ captures investment-specific technical change.
- Capital evolves according to:

$$\dot{K}_t = I_t - \delta K_t.$$

Equilibrium Prices

Model in Investment Units

- Investment good as the numeraire.
- Equilibrium relative prices are pinned down by sectoral productivities.

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}.$$

- Prices relative to investment are:

$$P_{g,t} = \frac{\mathcal{A}_t}{A_{g,t}}, \qquad P_{s,t} = \frac{\mathcal{A}_t}{A_{s,t}},$$

where

$$\mathcal{A}_t = A_{x,t}\left[\omega A_{g,t}^{\varepsilon-1} + (1-\omega)A_{s,t}^{\varepsilon-1}\right]^{\frac{1}{\varepsilon-1}}.$$

Aggregate Technology Representation

Aggregation Result

- Aggregate final output, measured in units of investment, is:

$$Y_t = \underbrace{P_{g,t}C_{g,t} + P_{s,t}C_{s,t}}_{E_t = \text{consumption expenditure}} + I_t$$

- Since factor ratios are equalized across sectors:

$$Y_t = \mathcal{A}_t K_t^{\theta} L_t^{1-\theta}.$$

$\mathcal{A}_t$ is aggregate TFP measured in units of investment.

Structural transformation is consistent with an aggregate production function.

Preferences and Structural Transformation

PIGL Demand System

- Preferences over g and s are represented by the PIGL indirect utility function:

$$V(e_t, P_{g,t}, P_{s,t}) = \frac{1}{\chi}\left[\left(\frac{e_t}{P_{s,t}}\right)^{\chi} - 1\right] + \frac{\eta}{\gamma}\left[1 - \left(\frac{P_{g,t}}{P_{s,t}}\right)^{\gamma}\right],$$

with

$$\eta > 0, \qquad 1 > \gamma > \chi > 0.$$

- Goods share in total consumption expenditure:

$$s_{g,t} = \frac{P_{g,t}c_{g,t}}{e_t} = \eta \underbrace{\left(\frac{e_t}{P_{s,t}}\right)^{-\chi}}_{\text{income effect}} \times \underbrace{\left(\frac{P_{g,t}}{P_{s,t}}\right)^{\gamma}}_{\text{substitution effect}}$$

◂ NHCES Preferences

Aggregate Balanced Growth Path

ABGP: BGP for aggregate variables + ST within Consumption and Investment

- Assume sectoral productivities grow at constant rates:

$$A_{g,t} = A_{g,0} e^{g_g t}, \qquad A_{s,t} = A_{s,0} e^{g_s t}, \qquad \mathcal{A}_t = \mathcal{A}_0 e^{g_{\mathcal{A}} t},$$

with

$$g_{\mathcal{A}} > g_g > g_s.$$

- Along the ABGP, aggregate per-capita variables grow at:

$$g_k = g_e = g_y = \frac{g_{\mathcal{A}}}{1-\theta} + g_h.$$

- The model features aggregate growth in investment units.
- But relative prices and expenditure shares still change over time.

Divisia Growth Along the ABGP

Real GDP Growth

- Along the ABGP, the Divisia growth rate of real GDP is:

$$g_t^D = s_e \underbrace{\left[s_{g,t} g_{c_g} + (1 - s_{g,t}) g_{c_s,t}\right]}_{\substack{\text{Growth rate of real} \\ \text{consumption expenditure}}} + (1 - s_e) g_x .$$

  - s_e: share of consumption expenditure in GDP.
  - g_x: growth rate of investment.
  - $s_{g,t}$: share of goods in consumption expenditure.
- Since $s_{g,t}$ and $g_{c_s,t}$ change over time, and $g_{c_g} > g_{c_s,t}$:

$$\implies g_t^D \text{declines along the ABGP.}$$

The Divisia growth rate differs from the growth rate of GDP measured in investment units.

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Key Questions

- Do chained indices provide a welfare-based measure of real income growth?
- Are they preferred to fixed-based equivalent variation measures?

Bellman Representation of Preferences

Non-Homothetic and Time-Varying Preferences

- The Bellman representation of preferences is:

$$W(e, x; P_{g,t}, P_{s,t}, \nu_t) = \max_{\{e,x\}} \; V(e, P_{g,t}, P_{s,t}) + \nu_t x$$

$$\text{s.t.} \qquad e + x = m_t$$

- ν_t is the marginal value of capital: $\nu_t = v'(k_t)$.
- The Bellman representation is:
  - Quasi-linear $\rightarrow$ Non-Homothetic.
  - Time-varying $\rightarrow$ $\{P_{g,t}, P_{s,t}, \nu_t\}$ change over time.

Bellman Representation of Preferences

Indirect Utility and Expenditure Functions

- In equilibrium, consumption expenditure is:

$$e_t = \left(\nu_t P_{s,t}^{\chi}\right)^{\frac{1}{\chi-1}}.$$

- Given net income m and prices (P_g, P_s), the indirect utility function is

$$u(m, P_g, P_s; \nu) = V\Big(\left(\nu P_s^{\chi}\right)^{\frac{1}{\chi-1}}, P_g, P_s\Big) + \nu\left[m - \left(\nu P_s^{\chi}\right)^{\frac{1}{\chi-1}}\right].$$

- The associated expenditure function is

$$f(w, P_g, P_s; \nu) = \left(\nu P_s^{\chi}\right)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\Big(\left(\nu P_s^{\chi}\right)^{\frac{1}{\chi-1}}, P_g, P_s\Big)}{\nu}.$$

- Quasi-linearity in investment makes m and w enter linearly.

Equivalent variation Measure of Income

- Fisher-Shell Principle: Welfare comparisons require a common preference order.
- Let
  - t: date of the preference order,
  - τ: date whose welfare is evaluated,
  - z: date of reference prices.

$$\widehat{m}_{t,z,\tau} = f\left(u\left(m_{\tau}, P_{g,\tau}, P_{s,\tau}; \nu_t\right), P_{g,z}, P_{s,z}; \nu_t\right).$$

- $\widehat{m}_{t,z,\tau}$: income at prices z needed to attain the welfare of period τ, using preferences of t.

Proposition 3

Equivalent Variation vs Current Income

- Under the Bellman representation,

$$\widehat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau}\,\widehat{e}_{t,z},$$

where

$$B_{t,z,\tau} = \left(\frac{1}{\chi} - 1 - \frac{\widehat{s}_{g,t,\tau}}{\gamma}\right)\left(\frac{P_{s,z}}{P_{s,\tau}}\right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\widehat{s}_{g,t,z}}{\gamma}\right).$$

- The EV measure equals observed income plus a price/preference correction.
- The correction depends on hypothetical consumption reallocation gains under the base preference order.

t → preference date τ → welfare comparison date z → reference price date

Proposition 4

Growth Rate of the Equivalent Variation Measure

- The growth rate of the EV income measure is

$$\frac{d \log \widehat{m}_{t,z,\tau}}{d\tau} = \frac{m_\tau}{\widehat{m}_{t,z,\tau}} \left(g_\tau^D + \mathrm{dev}_{t,z,\tau} \right).$$

- The Divisia growth rate appears as a benchmark:

$$g_\tau^D = s_e \left[s_{g,\tau} g_{c_g} + (1 - s_g) g_{c_s,\tau} \right] + (1 - s_e) g_x .$$

- Deviations from Divisia growth arise because fixed-base welfare comparisons imply different expenditure and sectoral shares.

$t \rightarrow$ preference date $\quad \tau \rightarrow$ welfare comparison date $\quad z \rightarrow$ reference price date

Three Welfare-Based Indices

For any time $z \in [t_0, t_1]$, with reference time t_0 and current time t_1.

1. **Chained Fisher–Shell = Chained Divisia index**

$$\mathcal{D}_z = \int_{t_0}^{z} g_h^D \, dh.$$

   Local measure: we compare τ with $\tau - \Delta\tau$, adopting the preferences and prices of τ

2. **Current-base Fisher–Shell index**

$$\mathcal{P}_{t_1,z} = \log \widehat{m}_{t_1,t_1,z} - \log \widehat{m}_{t_1,t_1,t_0}.$$

   The past is evaluated using current preferences and prices (like a Paasche Index).

3. **Reference-base Fisher–Shell index**

$$\mathcal{L}_{t_0,z} = \log \widehat{m}_{t_0,t_0,z} - \log m_{t_0}.$$

   The future is evaluated using reference-time preferences and prices (Laspeyres Index).

Proposition 5

Current-Base and Reference-Base Fisher–Shell Indices

- At the base date, fixed-base Fisher–Shell growth coincides with Divisia growth:

$$\left.\frac{d\mathcal{P}_{t_1,z}}{dz}\right|_{z=t_1} = g^D_{t_1}, \qquad \left.\frac{d\mathcal{L}_{t_0,z}}{dz}\right|_{z=t_0} = g^D_{t_0}.$$

- Away from the base date, the indices diverge:

$$\frac{d\mathcal{P}_{t_1,z}}{dz} < g^D_z < \frac{d\mathcal{L}_{t_0,z}}{dz} \qquad \Rightarrow \qquad \mathcal{P}_{t_1,z} < \mathcal{D}_z < \mathcal{L}_{t_0,z}.$$

Current-base measures understate past growth.
Reference-base measures overstate it.

$t \rightarrow$ preference date $\quad \tau \rightarrow$ welfare comparison date $\quad z \rightarrow$ reference price date

Plan

Calibration

Externally Calibrated Parameters

θ	ρ	δ	n	g_h	$g_{\mathcal{A}}$	g_s	g_g	ζ
0.333	0.040	0.080	0.0091	0.0046	0.0130	0.0024	0.0097	0.0137

- θ, ρ, δ taken from HRV.
- Extend HRV data through 2023.
- Compute exponential growth rates of:
  - population (n),
  - aggregate labor services (g_h),
  - sectoral ($\{g_s, g_g\}$) and aggregate ($g_{\mathcal{A}}$) TFP, via growth accounting.
- ζ matches the gap between observed and the model-implied relative prices.

Calibration

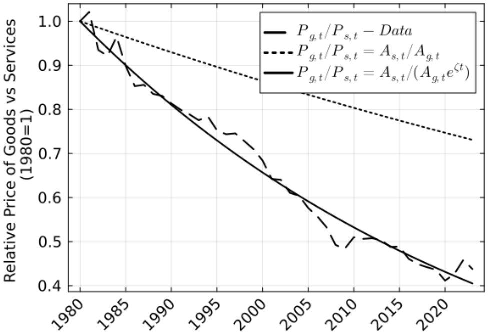
Preference Parameters Estimated via SMM

χ	η	γ
0.360	0.251	0.714

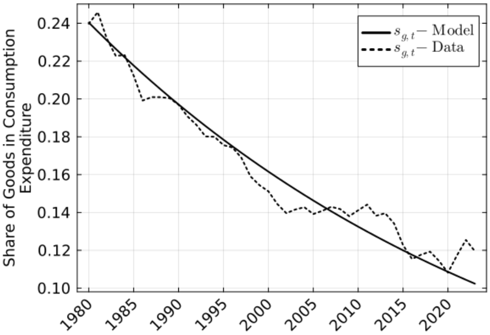
- η, χ, and γ are estimated by matching the goods' share in consumption expenditure, 1980–2023.
- The preference system captures both income effects and relative-price effects in consumption.

Calibration Fit

Relative Prices and Expenditure Shares



(a) Consumption goods relative price

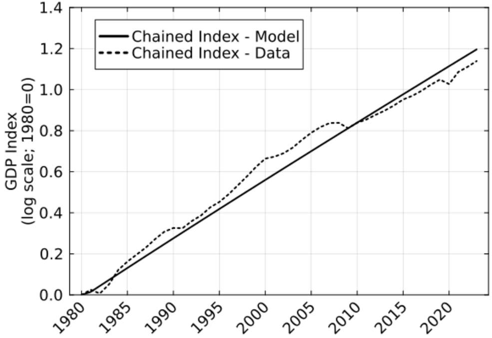


(b) Goods consumption share

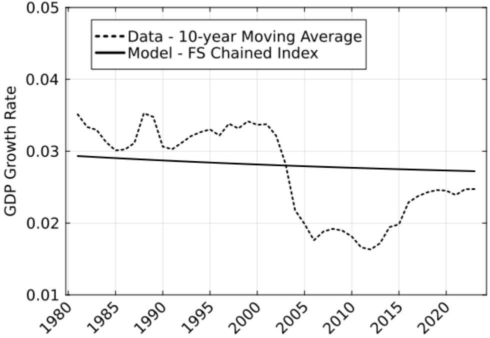
◂ Modeling the Wedge

Model Fit

GDP Level and Growth



(a) U.S. vs model GDP



(b) U.S. vs model GDP growth

◂ Model Fit Under NHCES

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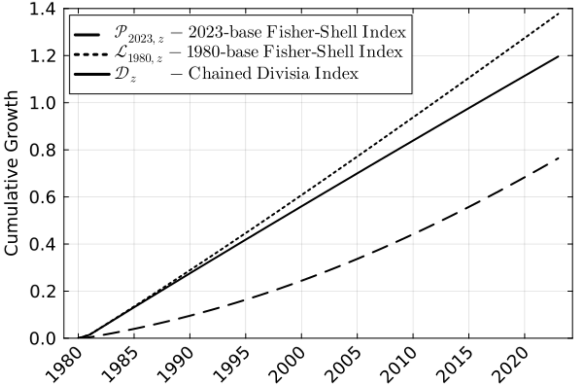
Welfare-Based Real GDP Measures

Fixed-Base Fisher-Shell Indices vs Chained Divisia Index

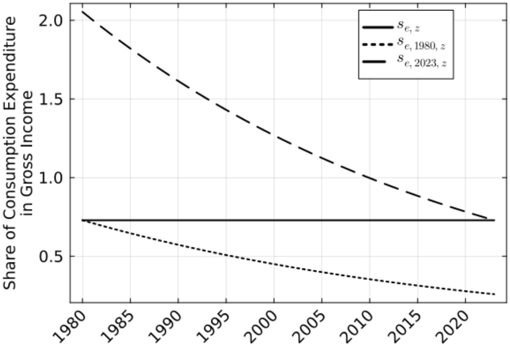


- The 1980-base Fisher–Shell index lies above the chained Divisia index (+18 p.p. in 2023).
- The 2023-base Fisher–Shell index lies below it (-43 p.p. in 2023).

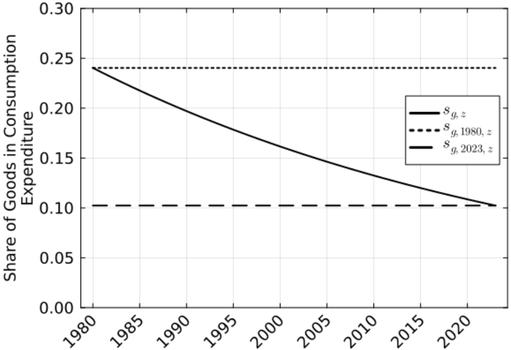
◂ Price-chained vs Pref.Chained Version ◂ Consumption Expenditure Indices ◂ Indices Under NHCES

Why Do the Indices Diverge?

Implied Expenditure Shares



(a) Consumption expenditure share



(b) Goods consumption share

- Fixed-base EV measures evaluate welfare using expenditure shares that differ from the observed equilibrium allocation.

◂ Exp. Shares Under NHCES

Economic Interpretation

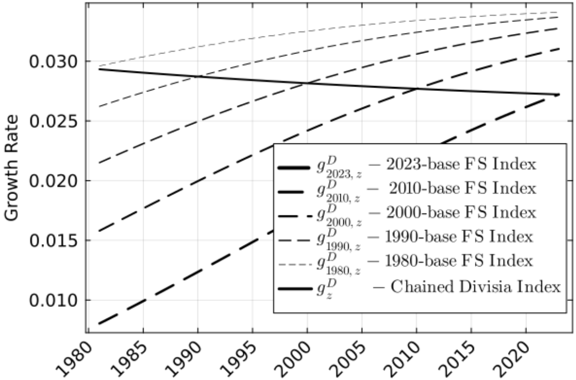
Base-Year Bias

- **1980-base Fisher–Shell index**
  - uses past preferences to evaluate later allocations;
  - assigns a declining weight to consumption;
  - therefore overweights investment, the faster-growing component;
  - overstates real income growth.
- **2023-base Fisher–Shell index**
  - uses current preferences to evaluate past allocations;
  - assigns an excessively high weight to consumption in the past;
  - therefore overweights the slower-growing component;
  - understates real income growth.

The chained Divisia index avoids these biases by updating prices and preferences locally.

Current-Time Upgrades

GDP History Revisions



- Repeated current-base evaluations systematically revise past growth downward.

◂ Growth Under NHCES

Plan

Conclusion

- Structural transformation and investment-specific technical change:
  - alter relative prices,
  - expenditure shares,
  - and the composition of output.
- Under these conditions:
  - welfare comparisons depend on the choice of preference base,
  - making fixed-base quantity indices systematically biased.
- The chained Divisia index:
  - continuously updates prices and expenditure weights,
  - coincides with the chained Fisher–Shell index in continuous time,
  - and provides a welfare-consistent measure of real income growth.
- Quantitatively:
  - fixed-base measures imply large and growing distortions,
  - and generate systematic revisions of historical growth.

Thank You

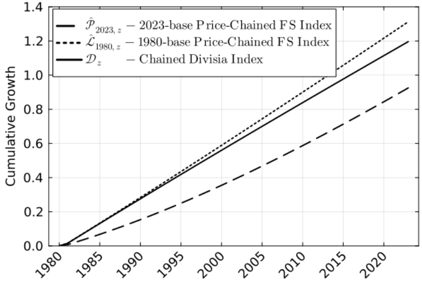
Replication files and online appendix available at:

`https://github.com/jivizcaino/Chained-Indices-Unchained`

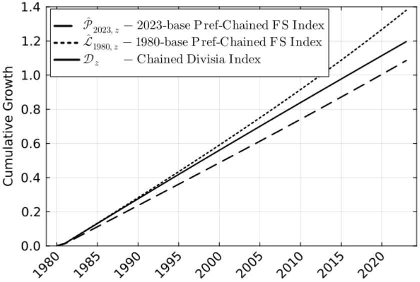
Appendix

Appendix

Price Chained and Preference Chained Fisher–Shell Indices



(a) Price-chained index



(b) Preference-chained index

(a) Prices update continuously, but preferences are kept fixed.
(b) Preferences update continuously, but prices are kept fixed

Most of the discrepancy with the Divisia index comes from holding preferences fixed.

◂ Back to Baseline Indices

Appendix

Indices for Real Consumption Expenditure

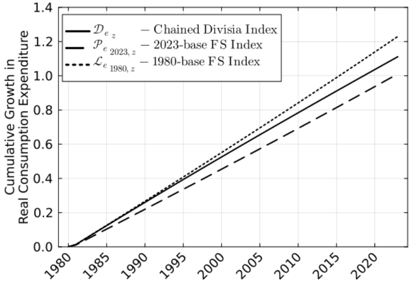
- In the endowment-economy analogue, the fixed-base Fisher–Shell growth rate for real consumption expenditure is

$$g_{e,t,z} = \frac{d \log \widehat{e}_{t,z}}{dz} = g^D_{e,z} \left(\frac{e_z}{\widehat{e}_{t,z}} \frac{P_{s,t}}{P_{s,z}} \right)^{\chi}.$$

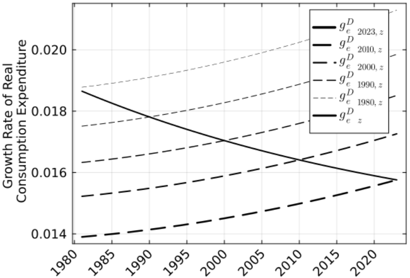
- A chained Fisher–Shell index again coincides with the chained Divisia index.
- Fixed-base indices still produce systematic base-year distortions under non-homotheticity.

Appendix

Real Consumption Expenditure Indices and Revisions



(a) Real consumption expenditure



(b) Growth-rate revisions

(a) Consumption expenditure explains part of the bias.
(b) Revisions are needed and fixed-based growth rates are increasing.

Most of the discrepancy with the Divisia index comes from the consumption-investment trade-off.

◂ Back to Baseline Indices

Appendix

Transformation Distortions in Consumption

- Household expenditure prices differ from producer prices:

$$P^{\tau}_{j,t} = e^{\tau_j t} P_{j,t}, \qquad j \in \{g, s\}.$$

- Producer prices and investment decisions are unchanged:

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}.$$

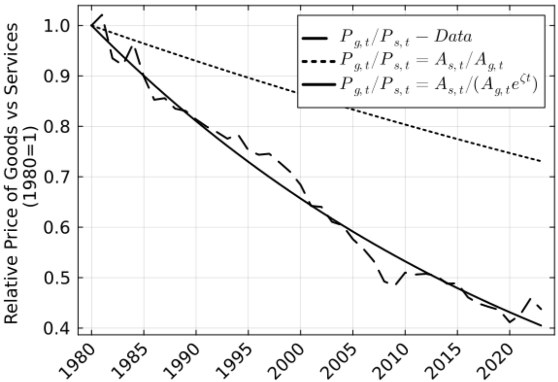
- Household relative expenditure prices become

$$\frac{P^{\tau}_{g,t}}{P^{\tau}_{s,t}} = \frac{A_{s,t}}{A_{g,t}} e^{(\tau_g - \tau_s)t}.$$

The wedge affects consumption allocation, but not production decisions, factor prices, or the aggregate technology representation.

Appendix

Transformation Wedge and Relative Prices



◂ Back to Calibration Fit

Appendix

- Replace PIGL demand with non-homothetic CES consumption:

$$\omega_c^{1/\sigma}\left(\frac{c_g}{C^{\eta_m}}\right)^{\frac{\sigma-1}{\sigma}} + (1-\omega_c)^{1/\sigma}\left(\frac{c_s}{C^{\eta_s}}\right)^{\frac{\sigma-1}{\sigma}} = 1.$$

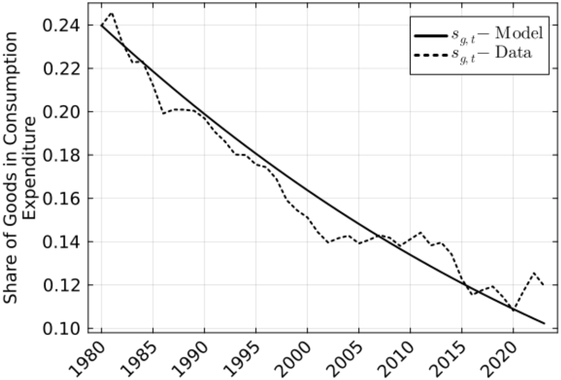
- The expenditure function is available in closed form:

$$E(P_g, P_s, C) = \left[\omega_c P_g^{1-\sigma} C^{(1-\sigma)\eta_m} + (1-\omega_c) P_s^{1-\sigma} C^{(1-\sigma)\eta_s}\right]^{\frac{1}{1-\sigma}}.$$

- The production side, investment block, factor prices, and aggregation result are unchanged.

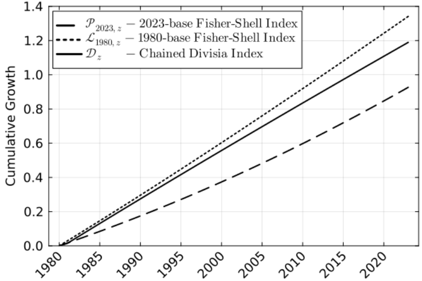
Appendix

NHCES Calibration Fit

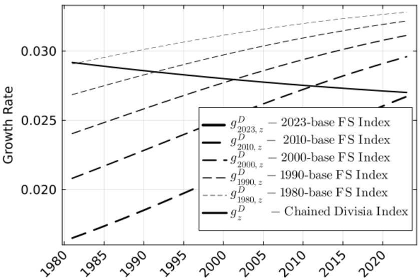


Appendix

Welfare-based GDP measures and revisions under NHCES



(a) Welfare-based GDP measures



(b) Revisions in growth rates

(a) The same qualitative pattern holds: reference-base FS lies above Divisia; current-base FS lies below it.

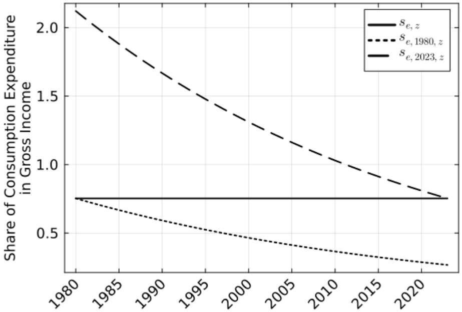
(b) Revisions are needed and fixed-base growth rates are increasing.

Similar quantiative discrepancy with the Divisia under NHCES preferences.

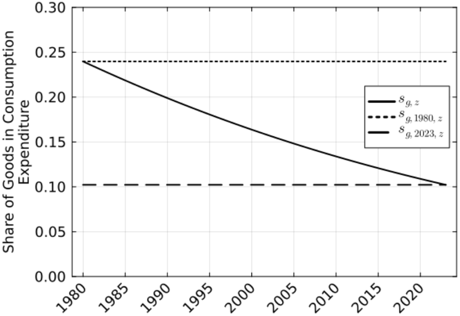
◂ Back to Baseline Indices

Appendix

Implied Expenditure Shares Under NHCES Preferences.



(a) Consumption expenditure share



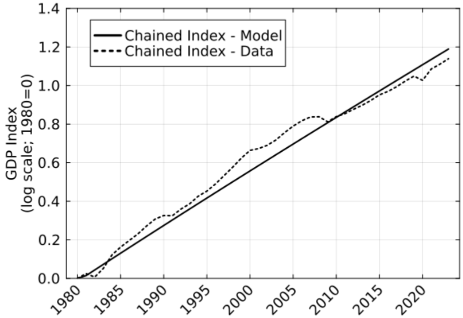
(b) Goods consumption share

- Hypothetical expenditure shares are quantiatatively similar under NHCES.

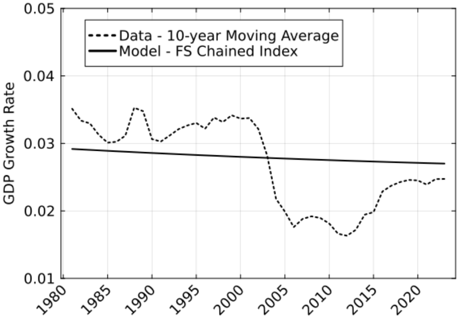
Back to Baseline Shares

Appendix

Model fit Under NHCES



(a) GDP index



(b) GDP growth

◂ Back to Model Fit